

Expertise

I am an AI researcher and engineer working in reinforcement learning, agentic systems, and large-scale training. As a founding-team member at Sakana AI, I built SOTA multi-agent systems, adaptive LLMs, and Kernel agent. I also have prior experience scaling multimodal foundation-model training on exascale supercomputers. My works are published in ICLR, AAAI, and NMI, and I enjoy working in small teams to tackle hard problems end-to-end.

Experience

Sakana AI	Member of Technical Staff
<i>Founding-team member building a Tokyo-based frontier AI lab from scratch.</i>	<i>Sep 2023–Present</i>

○ Multi Agent Orchestration.

- Co-developed [Sakana Fugu](#), a multi-agent system that achieves SOTA result on tasks such as SWE-Bench Pro, Terminal-Bench, and GPQA-D. Co-authored [Trinity](#) and [Conductor](#) papers, two core algorithmic foundations behind Fugu. I also lead the infrastructure serving on Google Cloud Platform for the API service.

○ Agentic AI Systems.

- Developed [Evolutionary Context Search](#), an evolutionary framework for agentic skill acquisition that searches over context combinations. ECS improves Gemini-3-Flash performance on BackendBench by 27%, and the discovered contexts transfer across models such as Claude-Sonnet.
- Co-developed [AI CUDA Engineer](#), the first open-source agentic system for CUDA kernel optimization, achieving 2x speedup over PyTorch on fused operations such as conv-relu-pool.

○ Adaptive Foundation Models.

- Co-developed [Transformer²](#), the first RL-trained self-adaptation framework for LLMs that infers task properties at inference time and mixes expert vectors to adapt model behavior across diverse tasks. It was accepted to ICLR, and the open-source release received 1.2K+ GitHub stars.
- Contributed to [Evolutionary Model Merging](#), an evolutionary approach that automatically discovers effective combinations of diverse foundation models without extensive training. The merged models achieved SOTA on several benchmarks at release, and the work was published in Nature Machine Intelligence.

Tokyo Institute of Technology	Research Assistant
<i>HPC × AI.</i>	<i>Sep 2022–Sep 2023</i>

○ Multimodal Foundation Models at Scale

- Set up large-scale training frameworks, including Megatron, DeepSpeed, and GPT-NeoX, across multiple exascale supercomputers, including Frontier (AMD MI250X), Jupyter(GH200), Summit (NVIDIA V100), scaling training up to 3K+ GPUs.
- Customized GPT-NeoX to support a vision encoder under pipeline parallelism, trained and released a scaling [suite](#) of vision-language models.
- Created an 800B-token synthetic dataset using the supercomputer in collaboration with LAION: [Game Dataset](#).

Education

Institute of Science Tokyo , Ph.D. in Computer Science	2024–Present
<i>Advisor: Rio Yokota</i>	
Tokyo Institute of Technology , M.S. in Computer Science	2022–2024
<i>Advisor: Rio Yokota</i>	
Dalian University of Technology , B.E. in Computer Science	2018–2022

Technical Skills

Machine Learning: RL, Agentic systems, Vision-language models, Multi-agent, Synthetic data generation,

Systems: Distributed training, Google Cloud Platform, SLURM, Kernel optimization

Programming: Python, Shell, PyTorch, JAX, Docker, Ray, CUDA, MPI.

Publications & Preprints

- Q. Sun, Nielsen, S., Yokota, R., & Tang, Y. *Evolutionary Context Search for Automated Skill Acquisition*. arXiv:2602.16113, 2026. **[Preprint]**
- Xu, J.^{*}, Q. Sun^{*}, Schwendeman, P., Nielsen, S., Cetin, E., et al. *TRINITY: An Evolved LLM Coordinator*. In Proceedings of the 14th International Conference on Learning Representations, 2026. **[ICLR 2026]**
- Nielsen, S.^{*}, Cetin, E.^{*}, Schwendeman, P.^{*}, Q. Sun, Xu, J., et al. *Learning to Orchestrate Agents in Natural Language with the Conductor*. In Proceedings of the 14th International Conference on Learning Representations, 2026. **[ICLR 2026]**
- Lange, R. T., Q. Sun, Prasad, A., Faldor, M., Tang, Y., et al. *Towards Robust Agentic CUDA Kernel Benchmarking, Verification, and Optimization*. arXiv:2509.14279, 2025. **[Preprint]**
- Q. Sun^{*}, Pickett, M.^{*}, Nain, A. K., & Jones, L. *Transformer Layers as Painters*. In Proceedings of the 39th AAAI Conference on Artificial Intelligence, 39(24), 25219–25227, 2025. doi:10.1609/aaai.v39i24.34708. **[AAAI 2025]**
- Q. Sun^{*}, Cetin, E.^{*}, & Tang, Y.^{*}. *Transformer²: Self-adaptive LLMs*. In Proceedings of the 13th International Conference on Learning Representations, 2025. **[ICLR 2025]**
- Cetin, E., Q. Sun, Zhao, T., & Tang, Y. *An Evolved Universal Transformer Memory*. In Proceedings of the 13th International Conference on Learning Representations, 2025. **[ICLR 2025]**
- Akiba, T., Shing, M., Tang, Y., Q. Sun, & Ha, D. *Evolutionary Optimization of Model Merging Recipes*. Nature Machine Intelligence, 7, 195–204, 2025. doi:10.1038/s42256-024-00975-8. **[NMI 2025]**
- Nakamura, T., Mishra, M., Tedeschi, S., Chai, Y., Stillerman, J. T., et al. *AURORA-M: Open Source Continual Pre-training for Multilingual Language and Code*. In Proceedings of the 31st International Conference on Computational Linguistics: Industry Track, 656–678, 2025. **[COLING 2025]**
- Roger, A.^{*}, Humane, P.^{*}, Kaplan, D. Z.^{*}, Gupta, K.^{*}, Q. Sun^{*}, et al. *CHIRP: A Fine-Grained Benchmark for Open-Ended Response Evaluation in Vision-Language Models*. arXiv:2501.09672, 2025. **[Preprint]**

^{*}Equal contribution.

Honors & Awards

2024–2026: Science Tokyo BOOST Scholarship, 10,800,000 JPY

2022–2023: Monbukagakusho Honors Scholarship, 300,000 JPY

2019–2020: Merit Student of Dalian University of Technology

2018–2019: China National Scholarship, 8,000 CNY